

Starting Finance Club PoliTo

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RESEARCH ARTICLE

Stochastic Valuation of Barrick Mining Corporation

An integrated option-implied and Monte Carlo valuation architecture

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Research collaboration

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Study design

Multi-team stochastic corporate valuation integrating market-implied gold-price dynamics with common operating and DCF assumptions.

Article scope

Condensed scientific-paper presentation of the integrated valuation architecture, findings, robustness and limitations.

JEL classification

G12; G13; G31; C32; C53; C61; C63

Abstract

This study develops and audits an integrated stochastic valuation framework for Barrick Mining Corporation within Starting Finance Club PoliTo Research. The architecture combines probabilistic modelling, econometric analysis, Monte Carlo simulation, historical gold-price dynamics, stochastic volatility and jump processes, option-implied calibration, purpose-built mine operating forecasts, and discounted cash flow valuation. Independent research modules are unified through explicit source ownership and consistent date, unit, measure, and accounting conventions. Alternative stochastic specifications are applied exclusively to the gold-price layer, while production and cost forecasts remain fixed across valuation engines through common operating-model inputs. Taxation, reinvestment, cost of capital, terminal value, and the equity bridge are likewise shared across scenarios. Option-market data are used to calibrate gold-price dynamics under the risk-neutral measure, allowing the effect of alternative distributional assumptions to be examined within an otherwise common corporate valuation contract. Results indicate that model choice can alter both central valuation outcomes and tail behaviour, while the observed market price remains above the medians generated by all four specifications. The resulting distributions are therefore interpreted as controlled stochastic valuation sensitivities rather than physical forecasts, confidence intervals, fair-value estimates, price targets, or investment recommendations.

1 INTRODUCTION

Valuing a gold producer under a single commodity-price path can conceal the model risk embedded in the commodity process itself. This paper is the publication-format endpoint of a broader eight-team research programme on Barrick Mining Corporation. The contributors came from different university pathways and therefore did not begin with the same exposure to measure-theoretic probability, econometrics, numerical simulation, software implementation, stochastic finance, and corporate valuation. For that reason, the project was first developed in a deliberately pedagogical, thesis-style form: definitions, derivations, diagnostics, implementation choices, empirical evidence and inter-module handoffs are made explicit rather than treated as common prior knowledge. The long-form version is intended to bridge these differences in background and provide a common technical language across the group. The present article takes the complementary editorial approach: it compresses that material into the methods and findings required to understand and audit the integrated valuation without reproducing the full teaching apparatus.

The eight source studies are therefore modular contributions, not eight replications of the same experiment. Teams 1–3 establish the mathematical, econometric and numerical foundations. **Team 1** develops the probability and pricing framework—measure and probability spaces, filtrations, Brownian motion, Itô calculus, martingales, change of measure, GBM and Black–Scholes—that fixes the notation and risk-neutral conventions used downstream. **Team 2** provides the econometric language for inference and time-series analysis through OLS and robust inference, stationarity, AR/MA/ARIMA dynamics, rolling dependence and DCC-type correlation models. **Team 3** develops the Monte Carlo layer, covering random-number generation, distributional transforms, path discretisation, estimator error, variance reduction and non-Gaussian dependence tools. Together, these studies supply the common analytical and computational vocabulary on which the later empirical modules rely.

Teams 6–8 progressively move from these foundations to the market-implied gold-price layer. **Team 6** studies the limitations of constant-

volatility continuous-path models and develops the jump and stochastic-volatility machinery—Poisson and compound-Poisson processes, Lévy ideas, characteristic functions, Fourier pricing, CIR variance, and the Merton, Heston and Bates benchmarks. **Team 7** provides the long-horizon empirical context for gold and silver, examining returns, dependence, heavy tails, volatility persistence, GARCH dynamics and long-memory diagnostics rather than treating the stochastic specifications as purely theoretical choices. **Team 8** then turns this stack into the current option-implied experiment: Black–Scholes, Heston, Bates–Poisson and affine Full Bates–Hawkes are calibrated to a structured GLD option surface using constrained nonlinear optimisation, tested through residual diagnostics and rolling one-step-ahead validation, and finally used to generate risk-neutral gold-path distributions.

The corporate side is supplied by Teams 4 and 5. **Team 4** builds the operating bridge from commodity prices to Barrick's business through mine-level production and cost modelling: log-space ARIMA dynamics for the common cost trend, SARIMA-based production components, bounded recovery-rate simulation and the resulting operating-margin/EBITDA inputs. **Team 5** supplies the valuation bridge from operations to enterprise and equity value through FCFF/FCFE logic, WACC, ROIC–growth consistency, terminal-value closure, scenario analysis and stochastic DCF. In the final experiment, these two modules are deliberately frozen so that the effect of changing the gold-price engine can be identified rather than confounded with simultaneous changes in production, costs or corporate assumptions.

The resulting architecture assigns each variable and methodological claim to a source authority and permits cross-module handoffs only when date, unit, probability measure and accounting meaning are explicit. The research question is therefore narrow and testable: how much does Barrick's conditional equity-value distribution change when the gold-price layer is replaced by progressively richer stochastic specifications while the operating and corporate layers remain fixed? Model assessment combines in-sample option calibration with one-step-ahead tests against an expanding mean and the previous-day implied-volatility surface, separating structural richness from predictive dominance.

2 DATA AND VALUATION SETUP

2.1 Controlled layer separation

The integrated experiment is organized as a controlled handoff across three modules. Team 8 governs only the gold-price process; Team 4 supplies the operating forecasts; and Team 5 maps those inputs through a common tax, WACC, terminal-value and equity-bridge contract. In the four-model valuation comparison, only the gold engine changes. Barrick's Q1–Q2 2026 actuals and the same Team 4 production and unit-cost forecasts from Q3 2026 onward are frozen over 20 quarters; the same Team 5 corporate rules and WACC shocks are then reused across runs.

Table 1. Frozen contract used for model attribution.

Block	Current experiment
Gold layer	Black–Scholes/GBM, Heston, Bates–Poisson and Full Bates–Hawkes calibrated by Team 8 under $\mathbb{Q}$.
Operations	Barrick Q1–Q2 2026 actuals followed by Team 4 deterministic production and unit-cost vectors from Q3 2026.
Corporate mapping	Team 5 tax, growth–ROIC–reinvestment schedule, stochastic WACC, terminal-value and equity-bridge logic.
Simulation	8,192 paths, 260 fine steps, 20 quarters and a common WACC shock array.

2.2 Calibration snapshot and measure discipline

The current Team 8 calibration uses the 27 August 2026 GLD option chain: 24 structured Chebyshev nodes across three maturities are selected from 143 eligible calls. The risk-free input is an NSS fit to the 24 July 2026 LSE US Treasury par-yield snapshot. Eleven tenors were returned; US10Y was not available, so it is not silently imputed as a raw observation. The fitted curve has a 2.37 bp RMSE and supplies maturity-matched continuously compounded rates. The Barrick market reference is the completed 27 August close of USD 47.27, while the common gold anchor is Barrick's Q2 2026 average realized price of USD 4,417/oz. The resulting date vector is explicit but asynchronous.

The option layer is estimated under the risk-neutral measure $\mathbb{Q}$. In the corporate experiment, only the option-implied distributional shape is transferred to the realized-price anchor. No one-for-one GLD-share-to-gold-ounce conversion and no validated $\mathbb{Q} \rightarrow \mathbb{P}$ transformation are claimed.

2.3 Operating-model status

Team 4's original operating study is richer than the object ultimately passed downstream. Cost of Sales is forecast from a log-space master chain; production is constructed bottom-up from ore processed, average grade and recovery. The unified valuation refreshes the handoff with 2026 actuals but still freezes the resulting 20-quarter production and cost vectors. Consequently, mine-level operating uncertainty is not propagated jointly with gold-price uncertainty. Team 4's legacy price simulation and illustrative EBITDA/valuation output are explicitly excluded.

3 METHODOLOGY

3.1 Gold-price engines

The four Team 8 specifications relax one restriction at a time. Under $\mathbb{Q}$, Black–Scholes/GBM uses constant volatility,

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sigma dW_t. \quad (1)$$

This constant-volatility specification also corresponds to the earlier internal working-paper draft that paired Team 4 operating forecasts with a provisional market-implied gold-price bridge based on GLD volatilities and a Nelson–Siegel–Svensson Treasury curve. In the present paper it is retained as the simplest benchmark inside the broader four-engine comparison. Heston introduces mean-reverting stochastic

variance (Heston, 1993),

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sqrt{v_t} dW_t^S, \quad (2)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \quad dW_t^S dW_t^v = \rho dt. \quad (3)$$

Bates adds compensated compound-Poisson jumps (Bates, 1996),

$$\frac{dS_t}{S_t} = (r_t - q_t - \lambda k) dt + \sqrt{v_t} dW_t^S + (e^J - 1) dN_t, \quad (4)$$

where $k = \mathbb{E}^{\mathbb{Q}}[e^J - 1]$.

3.2 Full Bates–Hawkes: implemented affine engine

The full engine preserves Heston variance but replaces the constant Poisson arrival rate with an exponential Hawkes state (Hawkes, 1971):

$$\frac{dS_t}{S_t} = (r_t - q_t - \lambda_t k) dt + \sqrt{v_t} dW_t^S + (e^J - 1) dN_t, \quad (5)$$

$$dv_t = \kappa(\theta_v - v_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (6)$$

$$d\lambda_t = \beta(\bar{\lambda} - \lambda_t) dt + \alpha dN_t, \quad n = \alpha/\beta < 1. \quad (7)$$

Under the implemented exponential-kernel specification, pricing exploits the affine Markov structure rather than a standalone finite-difference PIDE. Writing $X_t = \log S_t$, the characteristic function factorizes into Heston diffusion and Hawkes jump blocks,

$$\phi_{X_T}(u) = \phi_{\text{diff}}(u) \exp(A(u, T) + B(u, T)\lambda_0), \quad (8)$$

with the jump block determined by the Riccati system

$$\frac{dB}{d\tau} = \psi_J(u)e^{\alpha B} - \beta B - iuk - 1, \quad (9)$$

$$\frac{dA}{d\tau} = \beta \bar{\lambda} B, \quad A(0) = B(0) = 0, \quad (10)$$

where $\psi_J(u) = \exp(iu\mu_J - \frac{1}{2}\sigma_J^2 u^2)$. European option prices are recovered by Fourier-COS inversion. The identities $\phi(0) = 1$, the martingale condition at $u = -i$, and the limit $\alpha \rightarrow 0$ back to Bates are used as implementation checks.

Model-role constraint

These four processes govern only the gold-price layer. They never generate Barrick production or costs in the integrated valuation.

3.3 Option-surface calibration and predictive test

Every model is fitted to the same structured GLD surface by minimizing a vega-scaled price loss,

$$F(\theta) = \frac{1}{N} \sum_{i=1}^N \left[\frac{C_i^{\text{model}}(\theta) - C_i^{\text{mkt}}}{\max(\text{Vega}_i, \varepsilon)} \right]^2, \quad (11)$$

which approximates an implied-volatility error without solving an IV inversion inside every optimizer step. Richer models use global Differential Evolution for initialization and constrained SLSQP for local refinement. Feller admissibility is imposed on the square-root variance process, while Hawkes stability requires $n < 1$.

Calibration fit and forecasting are evaluated separately. One-step-ahead performance is summarized by

$$R_{OOS, m|b}^2 = 1 - \frac{\sum e_{m,t,j}^2}{\sum e_{b,t,j}^2}, \quad (12)$$

against both a node-specific expanding mean and the previous-day IV surface. This design prevents the lowest in-sample IV error from being interpreted automatically as predictive dominance.

3.4 Team 4 cost and production models

For costs, Team 4 first builds a log-space global master chain from mine-level series and then selects an ARIMA(3,1,0) with drift by automated information-criterion search. The admitted specification is

$$(1 + 0.9501B + 0.6128B^2 + 0.3003B^3)(1 - B)y_t = 0.0222 + \varepsilon_t. \quad (13)$$

The selected model minimizes both $AIC = -94.64$ and $BIC = -86.87$ (Akaike, 1974; Schwarz, 1978), where

$$AIC = 2k - 2 \log \hat{L}, \quad BIC = k \log n - 2 \log \hat{L}. \quad (14)$$

Residual diagnostics do not reject white noise or normality in the reported tests, and a 12-period hold-out yields a MAPE of 0.603% with RMSE approximately one third of the naive random-walk benchmark.

The production specifications are not reported as AIC/BIC-selected in the underlying Team 4 study. Instead, production is decomposed physically as

$$Q_t = \text{Ore}_t \text{ Grade}_t \text{ Recovery}_t \times 0.91/31.11, \quad (15)$$

with ore processed modelled as $SARIMA(1,0,1)(1,0,0)_4$, average grade as $SARIMA(1,0,0)(0,0,0)_4$, and recovery sampled from bounded historical ranges. The source study also develops Goodman variance propagation, but those operating distributions are not carried into the unified four-engine valuation; only the frozen point vectors are.

3.5 Valuation mapping

Team 5 makes growth internally consistent with reinvestment through

$$b_t = \frac{g_t}{ROIC_t}, \quad FCFF_t = M_t^{AT} - \max(M_t^{AT}, 0)b_t, \quad (16)$$

where M_t^{AT} is the after-tax component margin. Over the five explicit years, g_t declines linearly from 8.50% to 2.50%, $ROIC_t$ from 14.50% to 10.50%, and the implied reinvestment rate from 58.62% to 23.81%.

The common discount-rate process is mean reverting,

$$w_{t+1} = \bar{w} + (w_t - \bar{w})e^{-\kappa_w} + \sigma_w^* \varepsilon_{t+1}, \quad (17)$$

with $w_0 = 9.10\%$, $\bar{w} = 8.30\%$, $\kappa_w = 0.45$ and annual volatility 0.90%. It is clipped to $[g_\infty + 1.00\%, 25.00\%] = [3.50\%, 25.00\%]$; every gold engine receives the same 8,192-path shock array.

For simulated path i , the common operating contract and discount rates determine enterprise value,

$$EV^{(i)} = \sum_{t=1}^T \frac{FCFF_t^{(i)}}{\prod_{s=1}^t (1 + WACC_s^{(i)})} + \frac{TV_T^{(i)}}{\prod_{s=1}^T (1 + WACC_s^{(i)})}. \quad (18)$$

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and therefore $b_\infty = 23.81\%$:

$$TV_T^{(i)} = \frac{\max(M_T^{AT,(i)}, 0)(1 + g_\infty)(1 - g_\infty/ROIC_\infty)}{WACC_T^{(i)} - g_\infty}, \quad (19)$$

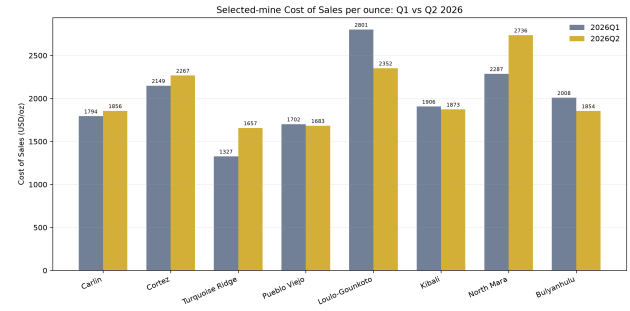
admitted only when $WACC_T^{(i)} > g_\infty$. The present component margin, FCFF and equity bridge remain unreconciled proxies; the final distributions are therefore conditional research sensitivities rather than filing-reconciled fair values.

4 CURRENT OPERATING-MODEL EVIDENCE

The cost and production layers inherited from Team 4 are purpose-built operating forecasts rather than generic statistical add-ons. They were first assembled in an earlier Barrick working paper that used a provisional Black–Scholes/GBM gold-price block. In the unified valuation only the operating method is retained and refreshed with Barrick's Q1–Q2 2026 mine statistics; the legacy gold-price simulation and illustrative EBITDA/valuation are excluded. This keeps the non-price inputs observable while ensuring that all valuation gold paths come from the current Team 8 engines.

4.1 Cost-of-sales block

The cost methodology stabilizes noisy mine-level CoS histories by extracting a common log-space market trend, forecasting that aggregate signal with the selected $ARIMA(3,1,0)$ with drift, and reconstructing mine paths with volatility-scaled anchors. AIC and BIC select the aggregate specification at -94.64 and -86.87 , respectively. Reported residual tests do not reject white noise or normality, while the 12-period hold-out produces a MAPE of 0.603% and an RMSE approximately one third of the naive random-walk benchmark. These diagnostics support the model-selection step; they do not turn CoS into a jointly simulated state in the final valuation.



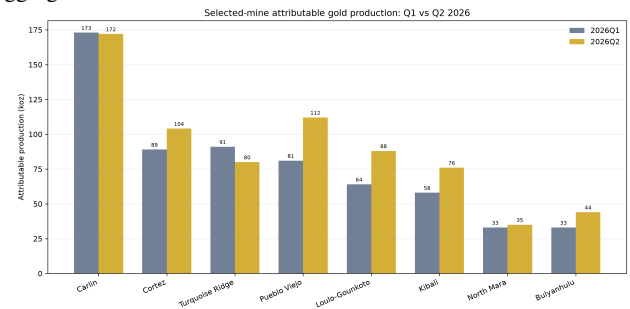
Source: Barrick Q1/Q2 2026 Mine Statistics. Selected Team 4 mine perimeter; company totals also include other operations.

Figure 1. Selected-mine Cost of Sales per ounce in Q1 and Q2 2026. Mine dispersion motivates preserving the Team 4 operating decomposition rather than replacing it with a single company-average cost.

The refreshed actuals provide a transparent starting boundary for the operating forecast. From Q3 2026 onward, the unified experiment accepts the resulting deterministic 20-quarter cost vector rather than the fully stochastic CoS distributions explored in the source study. This design is intentional: simultaneous changes in gold, cost and production uncertainty would prevent clean attribution across the four price engines.

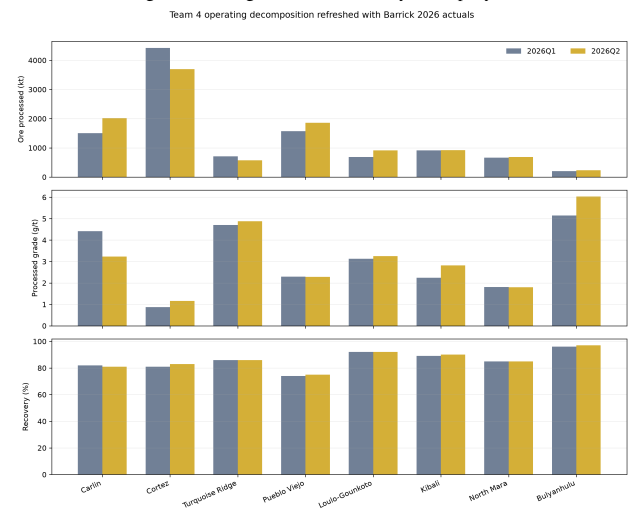
4.2 Production block

Production is modelled bottom-up rather than through a single total-output regression. Team 4 decomposes ounces into ore processed, average grade and recovery, using $SARIMA(1,0,1)(1,0,0)_4$ dynamics for ore, $SARIMA(1,0,0)(0,0,0)_4$ for grade and bounded stochastic simulation for recovery. The source does not report AIC/BIC selection for those production orders, so they are treated as admitted source specifications rather than re-selected models. Their physical recombination preserves mine heterogeneity and avoids the instability of a one-shot aggregate fit.



Source: Barrick Q1/Q2 2026 Mine Statistics. Selected Team 4 mine perimeter; company totals also include other operations.

Figure 2. Selected-mine attributable gold production in Q1 and Q2 2026. The current actuals reveal heterogeneous changes that are hidden by a company-level total.



Ore processed, grade and recovery are physical operating drivers; no gold-price simulation enters this figure.

Figure 3. Ore processed, grade and recovery updated with Barrick 2026 actuals. These are physical operating drivers; no commodity-price simulation enters the panel.

The original study also develops Goodman variance propagation and confidence bands for the production components. Those distributions remain useful evidence on operating-model uncertainty, but they are not propagated jointly in the four-engine corporate experiment: only

the refreshed point vectors are passed downstream.

4.3 Forecast handoff used by the valuation

The final handoff keeps observed and modelled quantities distinct. Barrick's Q1–Q2 2026 actual production and CoS establish the current operating state; the Team 4 forecasts begin at Q3 2026 and extend through the common valuation horizon.

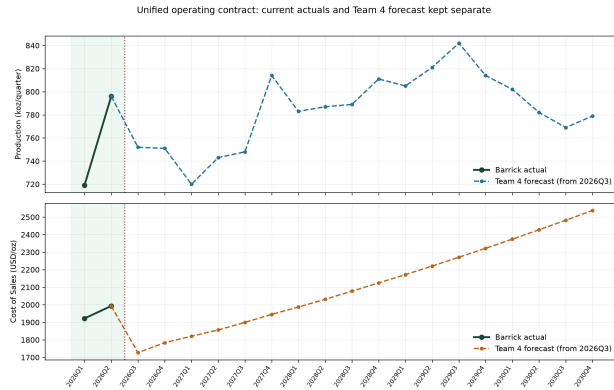


Figure 4. Unified operating contract. Barrick Q1–Q2 2026 actuals are shown separately from the deterministic Team 4 production and CoS vectors used from Q3 2026 onward.

Operating boundary

The four valuation branches reuse the same refreshed production and cost vectors. Team 4's legacy gold-price and illustrative valuation outputs remain excluded, so differences in the final distributions can be attributed to the Team 8 gold-price process under a common Team 5 DCF contract.

5 CALIBRATION SAMPLE AUDIT

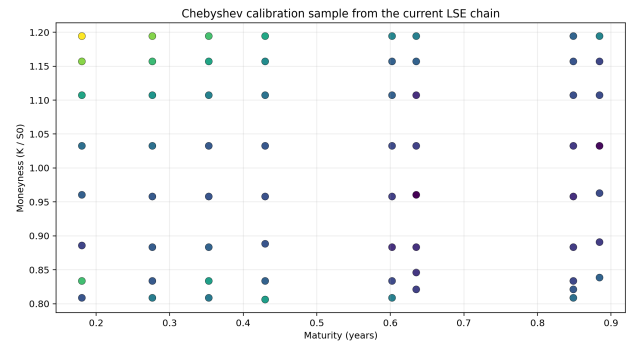


Figure 5. Chebyshev sampling geometry for the admitted 27 August 2026 GLD calibration: 24 retained nodes from 143 eligible calls across three maturities. Colour represents observed implied volatility.

5.1 Why structured sampling is retained

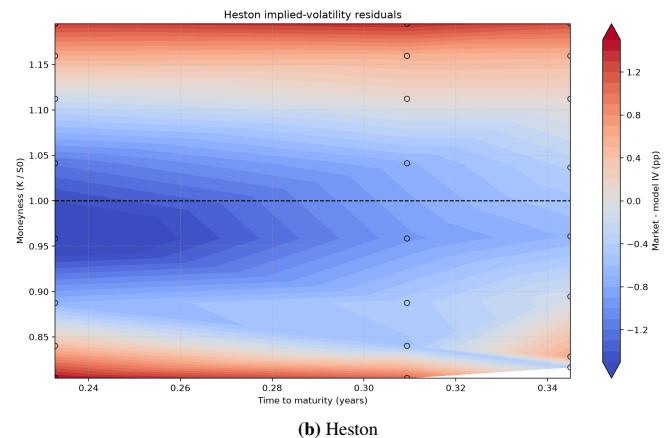
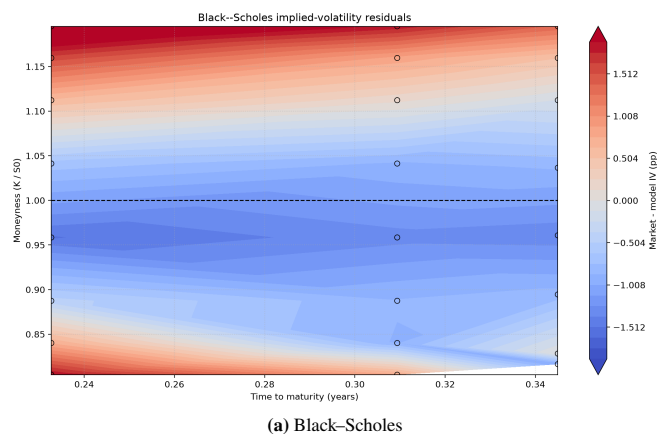
The option surface is irregular in strike and maturity. Team 8 therefore passes a structured Chebyshev subset to the optimizer rather than allowing densely listed regions to dominate the loss. The figure is the sampling audit for the same 24-node surface used by the current valuation, not a later diagnostic snapshot.

Evidence status

The calibrated inputs, residual maps and valuation paths share the same 27 August Team 8 parameter set. The Treasury curve remains dated 24 July and the timing mismatch is retained as an explicit limitation.

5.2 What the residual maps add

The residual heatmaps show where each specification systematically over- or under-prices the observed implied-volatility surface. Black–Scholes leaves the strongest smile and term-structure pattern; Heston reduces part of the curvature; jump-based specifications flatten the field further. Full Bates–Hawkes provides the lowest admitted IV RMSE, although its fitted branching ratio of 0.02 implies only mild self-excitation in this snapshot. The four maps are arranged as a two-by-two comparison so their common axes and colour scales remain directly comparable at a readable size.



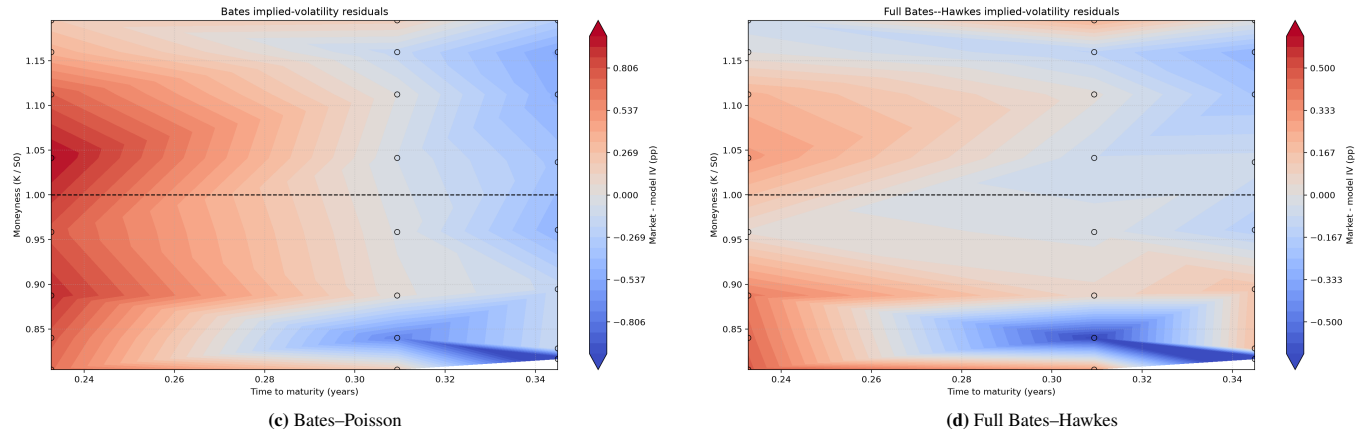


Figure 6. Current implied-volatility residual maps for Black-Scholes, Heston, Bates-Poisson and Full Bates-Hawkes. Blue denotes model IV above market IV and red underpricing. Model richness progressively flattens the residual field; the lowest in-sample IV RMSE belongs to Full Bates-Hawkes.

6 DCF DRIVER AND TERMINAL-VALUE AUDIT

The corporate layer is common to every gold engine, but it is not a black box. Growth, return on invested capital, reinvestment, discount rates and terminal closure are explicit configuration fields. Showing them separately prevents the stochastic gold model from being mistaken for the source of every valuation assumption.

6.1 Growth, ROIC and reinvestment

Growth declines linearly from 8.50% in year 1 to the 2.50% stable rate in year 5. ROIC converges from 14.50% to 10.50% over the same horizon. Because reinvestment is defined as $b_t = g_t / ROIC_t$, the corresponding annual rates are 58.62%, 51.85%, 44.00%, 34.78% and 23.81%. This declining schedule converts the high-growth phase into a stable-growth terminal regime rather than applying the same reinvestment intensity indefinitely.

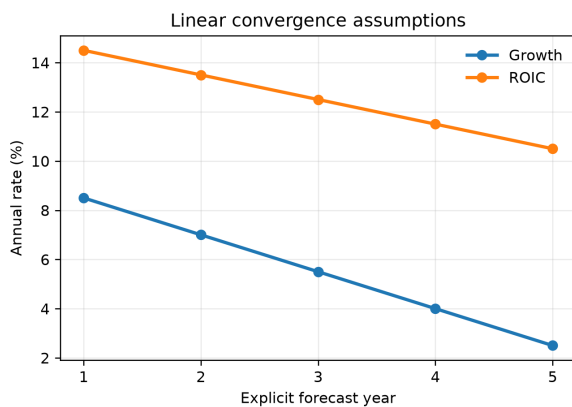


Figure 7. Common explicit-period DCF schedule: growth and ROIC converge linearly over the five-year horizon.

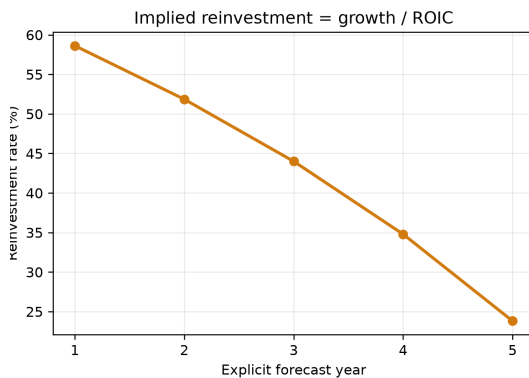


Figure 7 (continued). The implied reinvestment rate, defined as growth divided by ROIC, falls from 58.62% to 23.81%.

6.2 Common stochastic WACC

The WACC starts at 9.10% and mean reverts toward 8.30% with reversion speed 0.45 and annual volatility 0.90%. The annual conditional standard deviation used by the transition is 0.7308 percentage points. In the absence of shocks, the year-1 to year-5 path is 8.810%, 8.625%, 8.507%, 8.432% and 8.384%. Simulated values are clipped between 3.50%—the 2.50% stable growth rate plus a 1.00% spread floor—and 25.00%. The same 8,192-path shock array is reused in every valuation branch.

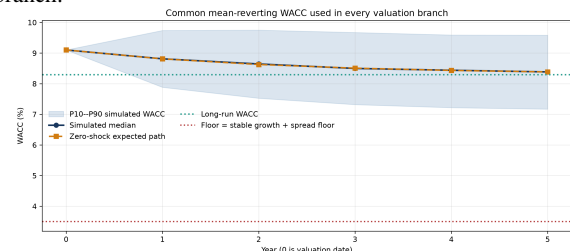


Figure 8. Common mean-reverting WACC. The band is the P10-P90 range from the shared 8,192-path shock array; the orange line is the zero-shock path and the dotted lines show the long-run level and admissibility floor.

6.3 Terminal closure and sensitivity

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and terminal reinvestment of 23.81%. At the current median terminal WACC of 8.384%, the normalized terminal multiple is 13.27 $\times$ final after-tax margin. The denominator is admitted only when $WACC_T > g_\infty$; the spread floor prevents a mechanically explosive closure in the simulated paths.

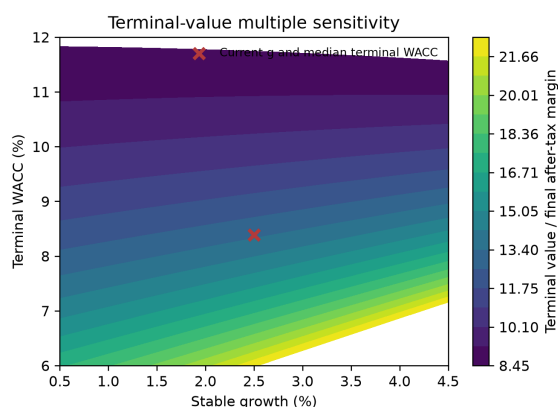


Figure 9. Terminal-value multiple sensitivity to stable growth and terminal WACC. The cross marks the configured stable-growth rate and current median terminal WACC.

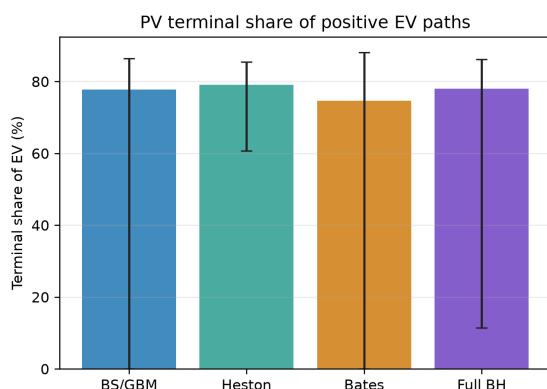


Figure 9 (continued). Present-value terminal share on positive-EV paths. Medians are 77.81% for Black–Scholes, 79.12% for Heston, 74.68% for Bates and 78.06% for Full Bates–Hawkes; bars show the P10–P90 range.

Table 2. Editable common DCF inputs.

Parameter	Value
Tax rate	25.00%
$g_1 \rightarrow g_\infty$	8.50–2.50%
$ROIC_1 \rightarrow ROIC_\infty$	14.50–10.50%
$WACC_0 \rightarrow \bar{w}$	9.10–8.30%
WACC reversion	0.45
WACC volatility	0.90%
WACC floor/cap	3.50/25.00%
Terminal reinvestment	23.81%

Interpretation

These are transparent methodological assumptions, not filing-calibrated Barrick estimates. Their materiality is visible: the terminal block supplies roughly three quarters of positive enterprise value. They should therefore be changed centrally and sensitivity-tested rather than hidden inside a single headline valuation.

7 CONDITIONAL VALUATION FINDINGS

Table 3. Primary Full Bates–Hawkes signal.

Metric	USD/share or %
P10	3.31
Median (P50)	34.94
P90	104.81
Completed market close	47.27
Market vs. median	+35.27%
Market percentile	62.27th
Paths above market	37.73%

Interpretation

The completed market close is above the Full Bates–Hawkes conditional median but remains inside a wide right-skewed distribution. The result therefore leans toward overvaluation relative to this model median; it does not establish market mispricing.

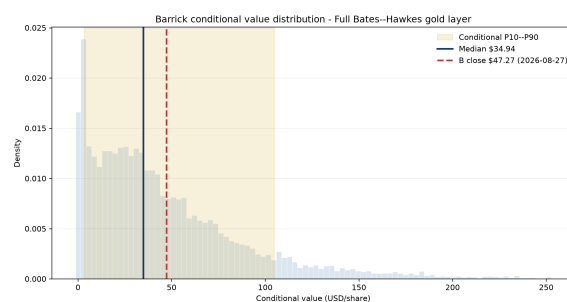


Figure 10. Conditional Barrick value-per-share distribution under the Full Bates–Hawkes gold layer. The reference line is the completed 27 August 2026 close. Current Team 4 operating vectors and Team 5 corporate assumptions are common across 8,192 paths; the band is not a target-price or confidence interval.

7.1 Cross-model central tendency

The updated run no longer supports a claim that all four conditional medians are tightly clustered. Black–Scholes, Heston and Full Bates–Hawkes lie between USD 34.50 and USD 39.48 per share, whereas Bates–Poisson has a lower median of USD 23.99 and a wider upper tail. Under the fixed operating and corporate contract, model choice therefore changes both central tendency and tail shape.

Table 4. Central comparison across gold engines.

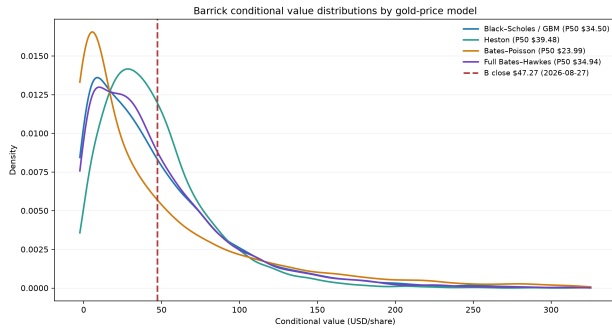
Gold engine	Median	Pr($V > 47.27$)
Black–Scholes	34.50	37.72%
Heston	39.48	40.06%
Bates–Poisson	23.99	33.89%
Full Bates–Hawkes	34.94	37.73%

7.2 Why Full Bates–Hawkes is primary

Full Bates–Hawkes is displayed as the primary structural scenario because it combines stochastic variance with self-exciting jumps and achieves the lowest in-sample IV RMSE on the current GLD surface. The ranking is monotonic in model richness, but the fitted Hawkes branching ratio is only 0.02. The surface therefore supports the richer specification as an admitted structural fit without implying a large self-excitation contribution or uniquely superior forecasting performance.

Table 5. Current-surface IV calibration RMSE.

Model	RMSE (bp)
Black–Scholes	114.04
Heston	86.28
Bates–Poisson	56.55
Full Bates–Hawkes	32.49



Only the Team 8 gold-price engine changes. The separated operating contract, unified DCF/WACC and unresolved corporate bridge are common.

Figure 11. Overlay of conditional Barrick value distributions under the four gold-price engines. The USD 47.27 close sits above every conditional median. Operations, WACC and the DCF mapping are common, so the separation of the Bates centre and right tail is attributable to its gold-path distribution rather than to a different company model.

8 OUT-OF-SAMPLE EVIDENCE AND MODEL RISK

Team 8: one-step-ahead predictive performance

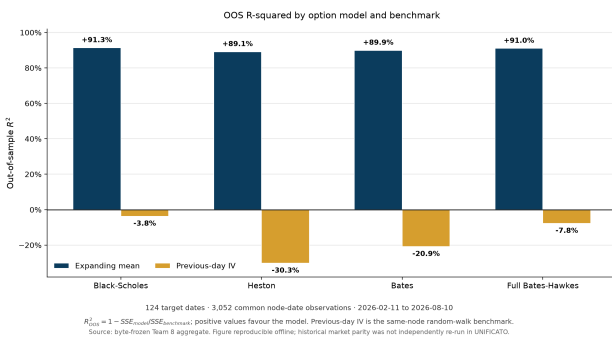


Figure 12. One-step-ahead out-of-sample R^2 . All four models improve on the expanding mean but fail to beat the previous-day IV surface. The test covers 124 target dates and 3,052 common node-date observations.

8.1 Predictive interpretation

The OOS evidence prevents the in-sample ranking from being read as predictive dominance. Relative to an expanding mean, the four models explain a large share of one-step-ahead variation. Against the stronger same-node previous-day IV benchmark, every R^2_{OOS} is negative; Black-Scholes also posts a slightly lower pooled RMSE than Full Bates-Hawkes. Structural flexibility therefore improves current-surface fit more clearly than short-horizon forecasting.

Model-risk conclusion

Full Bates-Hawkes is primary for structural richness and current-surface calibration quality, but the valuation conclusion is reported across all four engines because the OOS evidence does not identify a dominant forecasting model.

8.2 Interpretive limitations

The economic reading remains conditional: calibration is under $\mathbb{Q}$ without a validated $\mathbb{Q} \rightarrow \mathbb{P}$ map; GLD distributional shape is transferred without a one-for-one conversion to physical gold; market inputs are asynchronous; and the FCFF/equity bridge is not fully filing-reconciled. Debt, cash, minorities, non-operating assets and related bridge adjustments remain unresolved rather than silently inferred. The distributions are therefore research sensitivities, not fair values, confidence intervals or price targets.

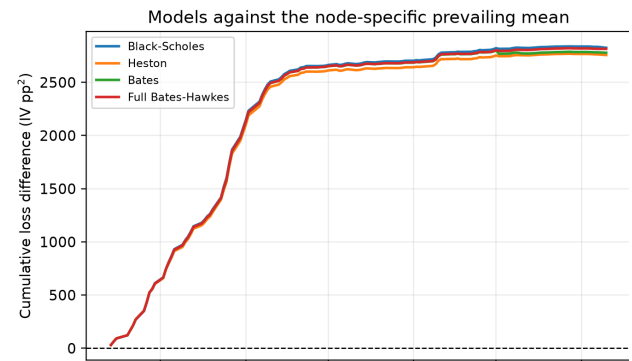


Figure 13. Rolling Welch-Goyal cumulative loss difference against the node-specific prevailing mean. Upward movement favours the named model; all four specifications dominate this benchmark.

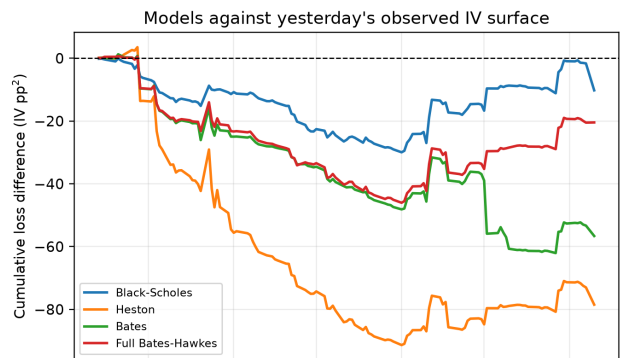


Figure 13 (continued). Cumulative loss difference against the previous-day observed IV surface. Negative paths show that this stronger short-horizon benchmark remains superior.

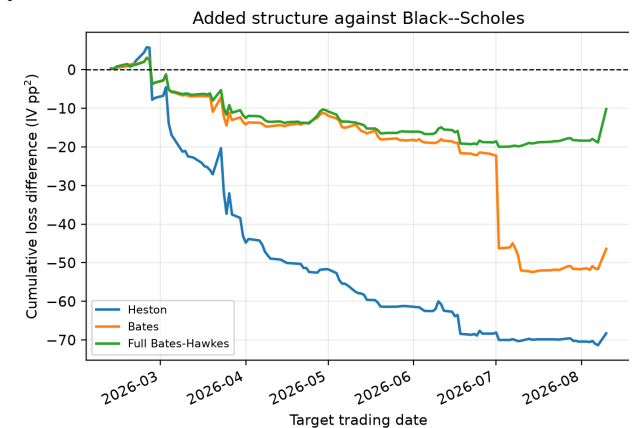


Figure 13 (continued). Added stochastic-volatility and jump structure relative to Black-Scholes. Negative paths indicate that the simpler benchmark retains lower cumulative loss over this test.

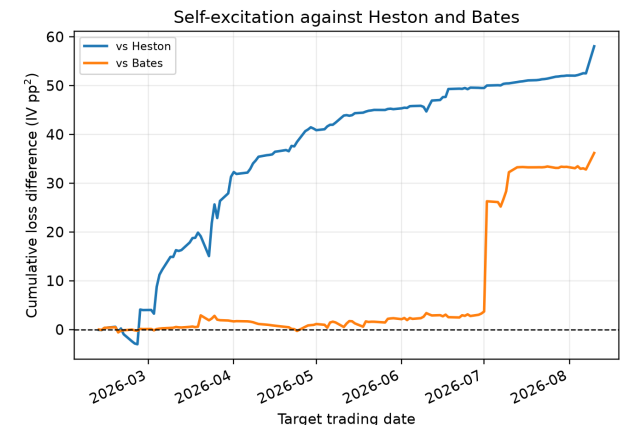


Figure 13 (continued). Full Bates-Hawkes relative to Heston and Bates. Positive paths favour the self-exciting specification within this pairwise comparison.

8.3 Jump-layer interpretation

The self-exciting extension is motivated by clustered option-implied jump activity. Relative to the constant Poisson baseline, the Hawkes specification permits bursts of intensity followed by endogenous decay. In the current calibration the branching ratio is 0.02, so this channel is present but economically mild.

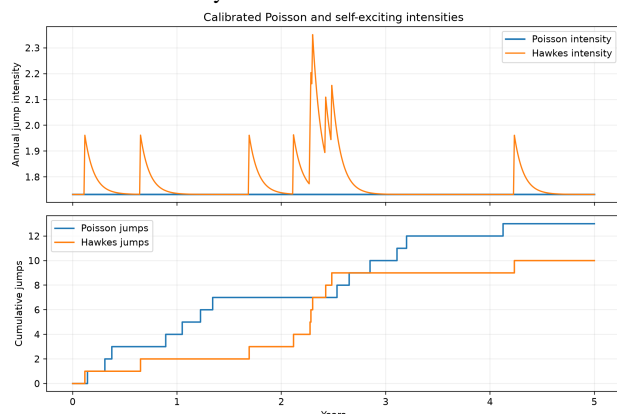


Figure 14. Current Poisson and Hawkes intensity comparison. Hawkes allows clustered arrivals and decay, but the low fitted branching ratio keeps it close to the constant-intensity benchmark in this snapshot.

9 CONCLUSION

The experiment isolates the effect of stochastic gold dynamics while holding operations and corporate assumptions fixed. Black–Scholes, Heston and Full Bates–Hawkes retain broadly comparable medians, while Bates–Poisson produces a lower centre and wider upper tail. The completed market close lies above every conditional median, a directional result narrower than a claim of market mispricing.

Model quality is multidimensional. Full Bates–Hawkes provides

the strongest current-surface fit, yet its fitted branching ratio is small and the rolling test does not establish universal forecasting superiority: none of the four models beats the previous-day IV benchmark. Team 4’s updated operating layer uses Q1–Q2 2026 actuals and a separate Q3-ongoing forecast, but its uncertainty is frozen before model comparison.

The paper should therefore be read as a controlled attribution exercise. The newly explicit growth, WACC and terminal-value assumptions make the DCF auditable, but the terminal block still contributes roughly three quarters of positive enterprise value. Any fair-value interpretation would additionally require a reconciled commodity basis, physical-measure mapping, operating uncertainty and equity bridge.

Decision-relevant takeaway

Under a common operating and DCF contract, changing the stochastic gold engine can affect both the central valuation signal and tail risk. The current architecture is informative for model comparison, but it is not yet a filing-reconciled fair-value model.

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